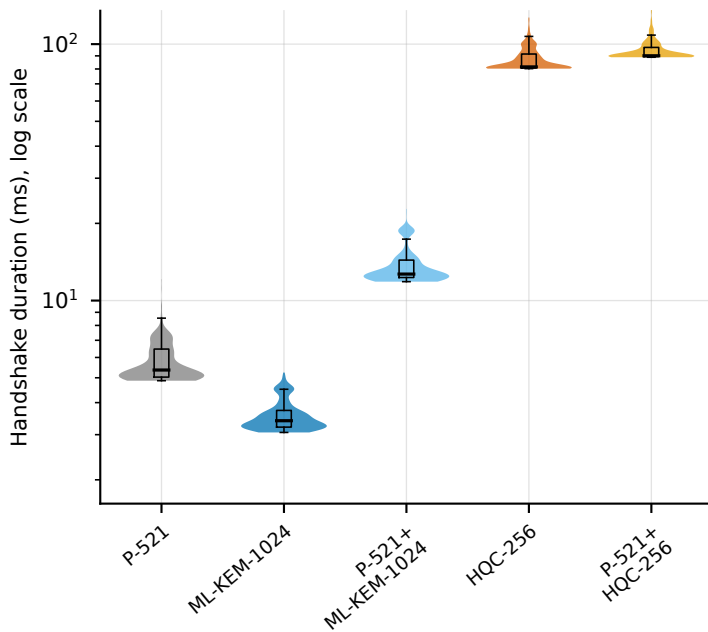
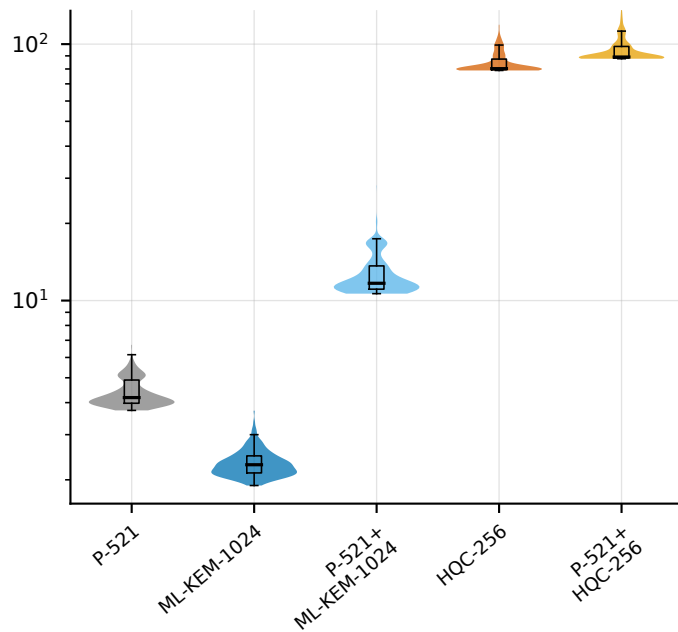


(a)

secp521r1 (classical)

**(b)**

mlds87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mlds87 Mean (ms)	mlds87 CV (%)	mlds87 Max (ms)
P-521	5.82	17.5	12.1	4.41	12.7	6.72
ML-KEM-1024	3.59	13.9	5.24	2.34	12.2	3.72
P-521+ML-KEM-1024	13.8	16.1	22.8	12.8	17.9	28.1
HQC-256	87.1	10.3	127.0	84.6	9.5	119.0
P-521+HQC-256	94.7	9.5	156.2	94.5	10.7	144.5